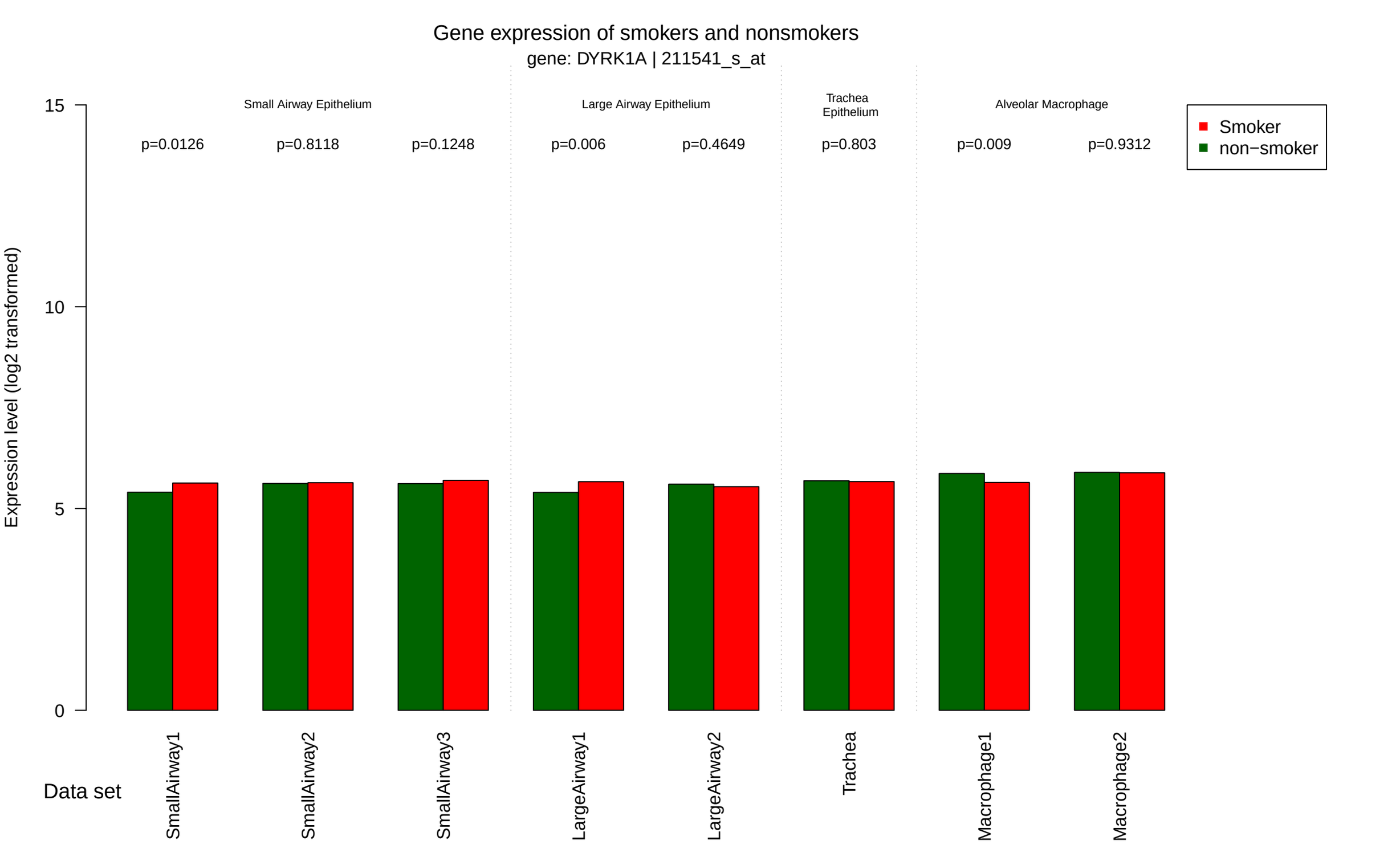
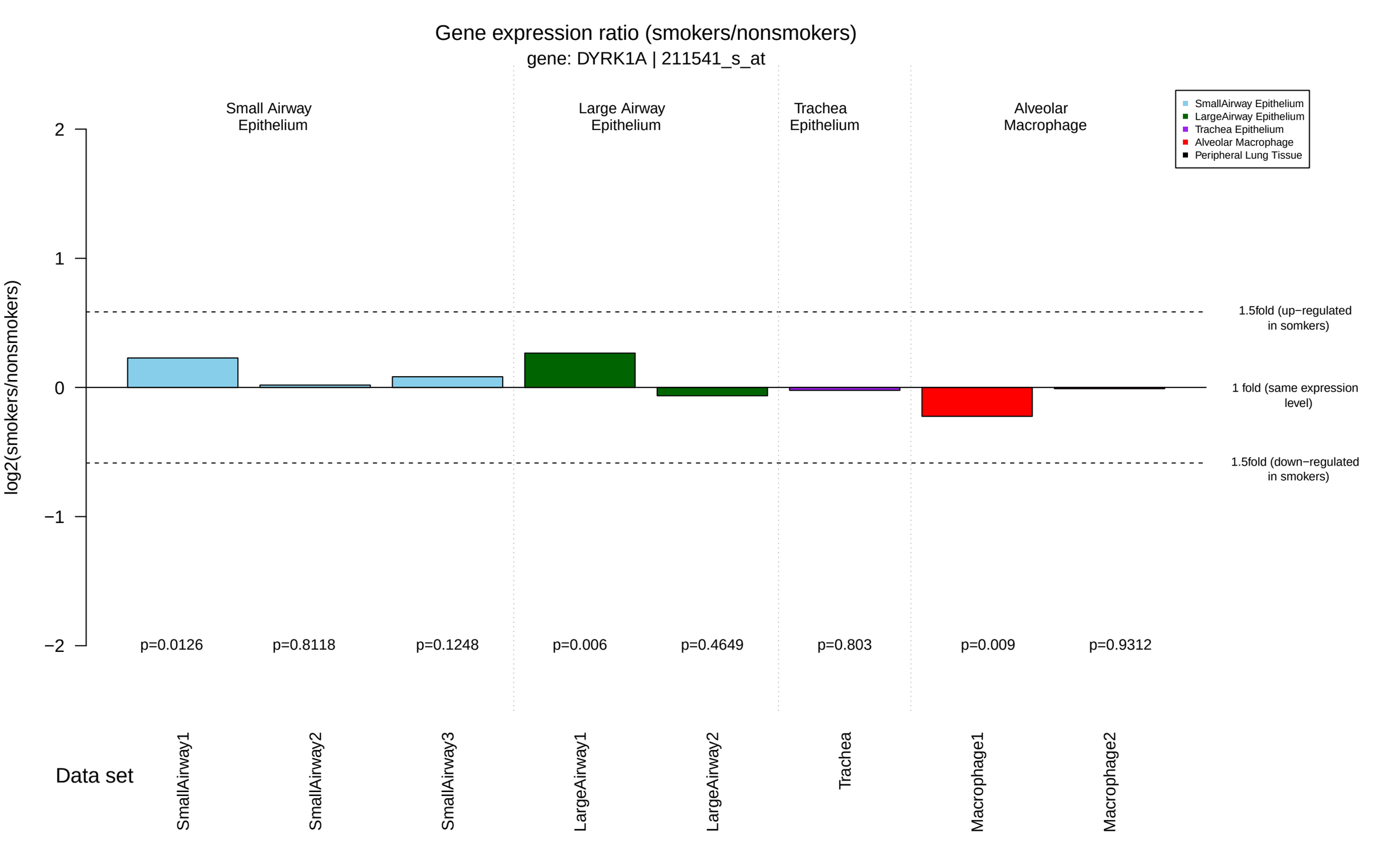
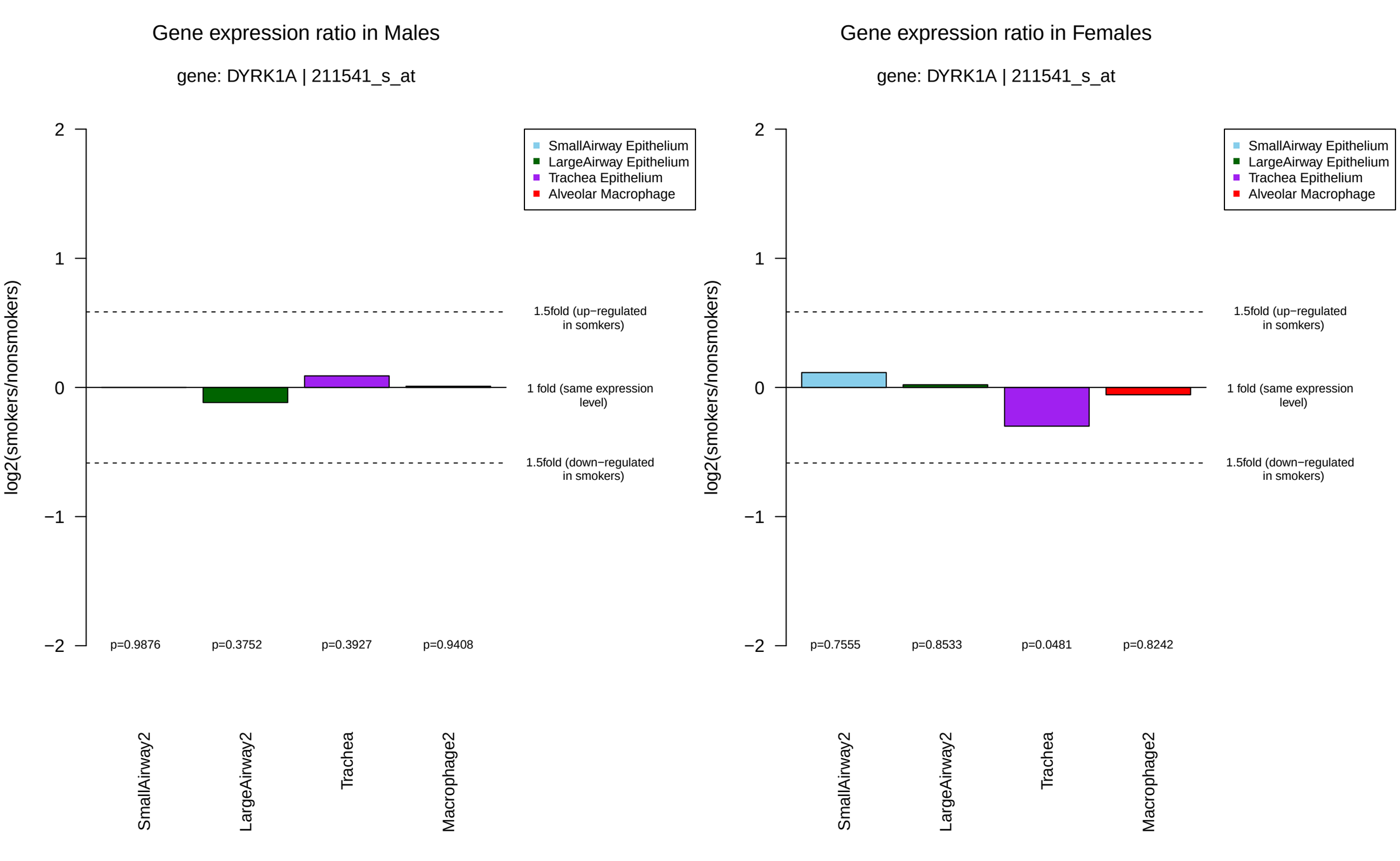
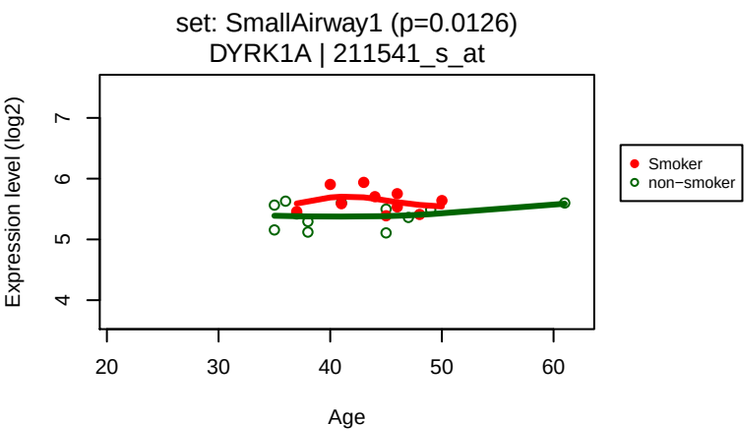
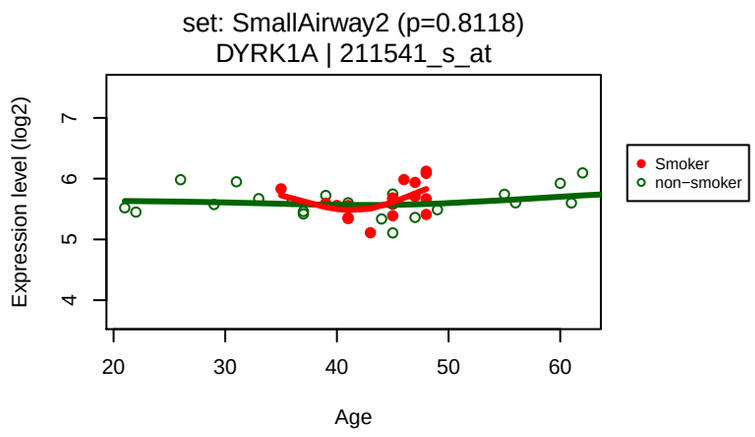
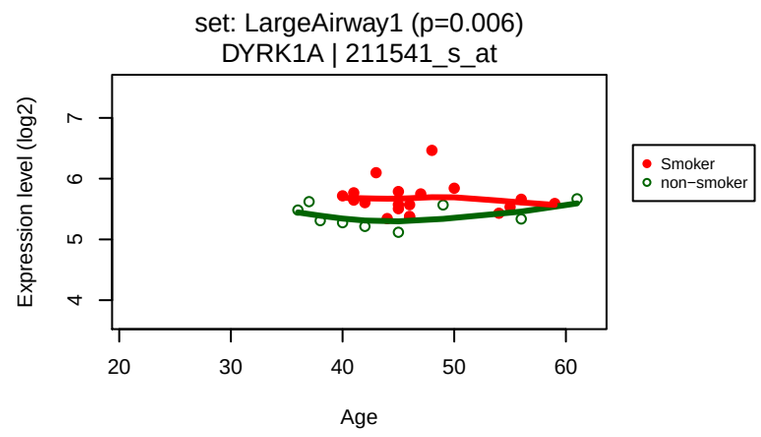

Gene expression of smokers and nonsmokers
gene: DYRK1A | 211541_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0126
p=0.8118
p=0.1248
p=0.006
p=0.4649
p=0.803
p=0.009
p=0.9312
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

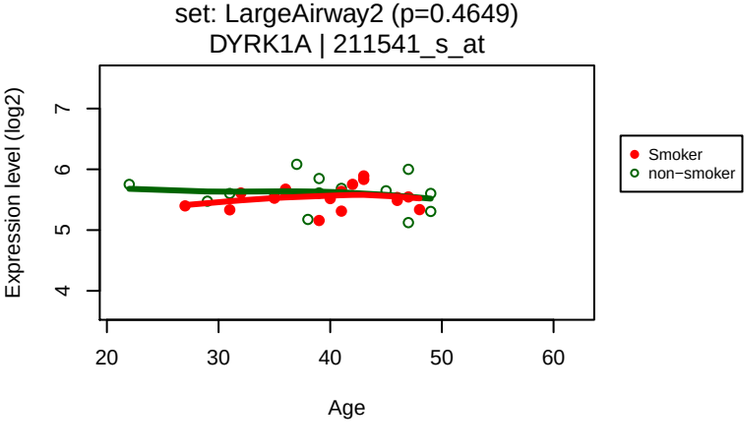

Gene expression ratio (smokers/nonsmokers)
gene: DYRK1A | 211541_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0126
p=0.8118
p=0.1248
p=0.006
p=0.4649
p=0.803
p=0.009
p=0.9312
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

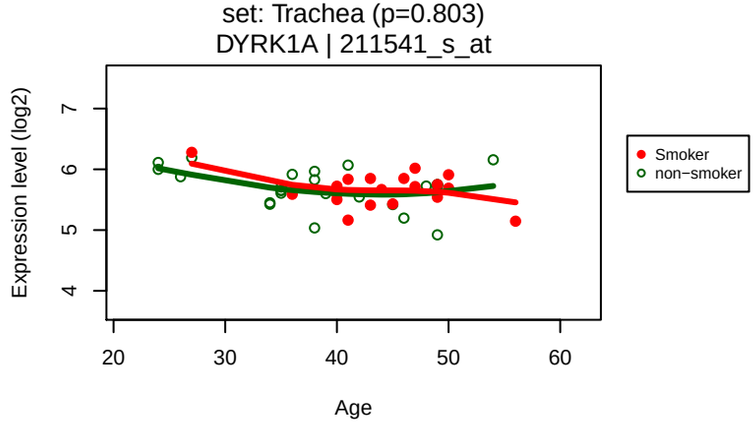

Gene expression ratio in Males
gene: DYRK1A | 211541_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9876
p=0.3752
p=0.3927
p=0.9408
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: DYRK1A | 211541_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7555
p=0.8533
p=0.0481
p=0.8242
SmallAirway2
LargeAirway2
Trachea
Macrophage2

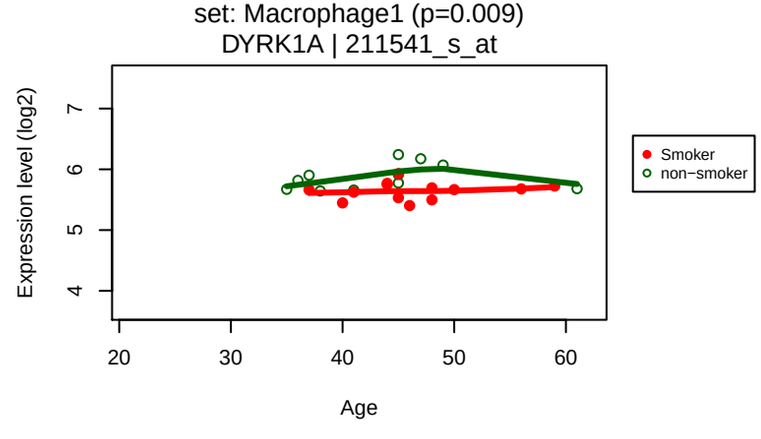

set: SmallAirway1 (p=0.0126)
DYRK1A | 211541_s_at
Expression level (log2)
Age
Smoker
non−smoker

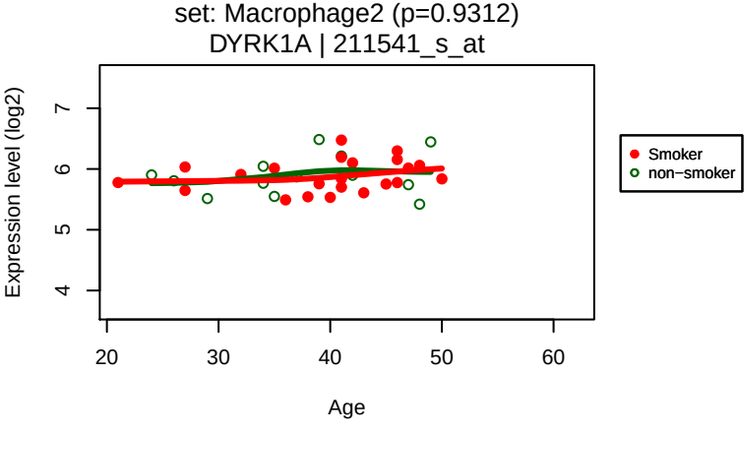

set: SmallAirway2 (p=0.8118)
DYRK1A | 211541_s_at
Expression level (log2)
Age
Smoker
non−smoker

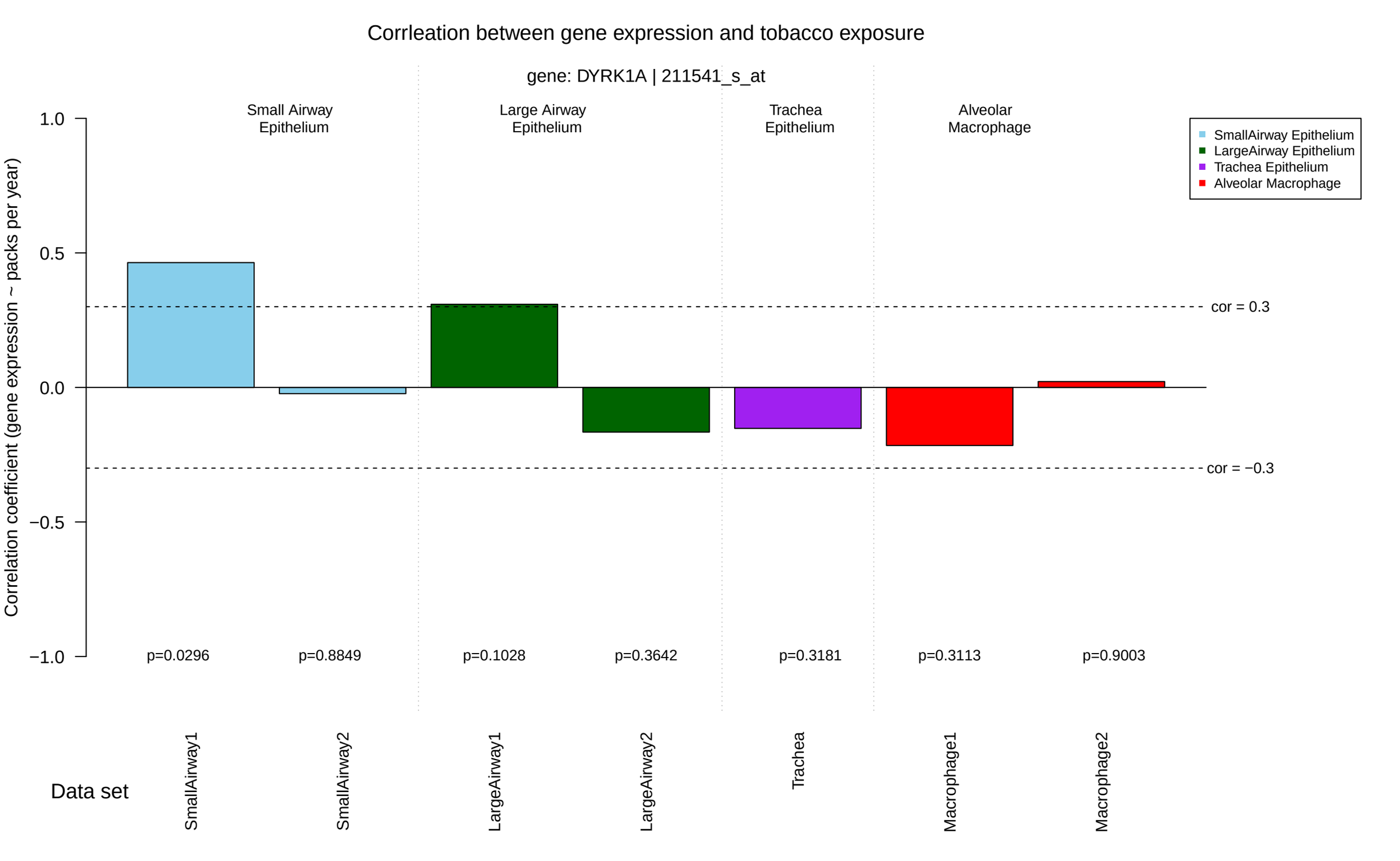

set: LargeAirway1 (p=0.006)
DYRK1A | 211541_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4649)
DYRK1A | 211541_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.803)
DYRK1A | 211541_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.009)
DYRK1A | 211541_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9312)
DYRK1A | 211541_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: DYRK1A | 211541_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0296
p=0.8849
p=0.1028
p=0.3642
p=0.3181
p=0.3113
p=0.9003
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2